Unit Fractions in Area Models
Answer Key

Grade 3 — Fractions as Numbers

Quick Answers

1. $1 \cdot \frac{1}{4}$
2. $1 \cdot \frac{1}{3}$
3. $1 \cdot \frac{1}{6}$
4. (a) the unit fraction shaded $= 1 \cdot \frac{1}{8}$, (b) how many of those pieces fill the whole $= 8$
5. —
6. $>$
7. $1 \cdot \frac{1}{8}$, $1 \cdot \frac{1}{4}$, $1 \cdot \frac{1}{2}$
8. (a) the unit fraction for one plot $= 1 \cdot \frac{1}{8}$, —

What is verified

6 of 8 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 5, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 3 — Fractions as Numbers
3.NF.A – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–5 of 5 across 11 tagged checks.

1. A rectangle is cut into 4 equal pieces; one piece is shaded. What unit fraction is shaded?
Solution: Count the equal pieces that make the whole rectangle: there are 4 of them. A unit fraction names one of those pieces, so the top number is 1 and the bottom number is the count of equal pieces. $\boxed{\frac{1}{4}}$

2. A circle is cut into 3 equal pieces; one piece is shaded. What unit fraction is shaded?
Solution: The whole circle is made of 3 equal wedges. One wedge out of those three is one third. The shape changed from a rectangle to a circle, but the rule did not: the bottom number is the number of equal pieces in the whole. $\boxed{\frac{1}{3}}$

3. A window pane is cut into 6 equal pieces; one piece is shaded. What unit fraction is shaded?
Solution: The pane is 2 rows of 3 panes, so the whole is 6 equal pieces. One of them is one sixth. Note that the pieces are arranged in a grid rather than a strip; what matters is that all six are the same size. $\boxed{\frac{1}{6}}$

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4. A chocolate bar is cut into 8 equal pieces; one piece is shaded. (a) What unit fraction is shaded? (b) How many pieces that size fill the whole bar?

Solution (a): 8 equal pieces make the whole bar, so one piece is one eighth. $\boxed{\frac{1}{8}}$

Solution (b): Ask how many one-eighths fit in 1 whole: $1 \div \frac{1}{8} = 8$. This is the same fact read backwards — a unit fraction's bottom number IS the number of copies it takes to rebuild the whole. $\boxed{8}$

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5. Two shapes are each cut into four pieces, one piece shaded. (a) Circle the model that shows fourths. (b) Explain why the other does not.

Solution (a) — instructor-judged. Model **B** is the one that shows fourths: its four pieces are all the same width. Model A also has four pieces, but they are different sizes, so no single piece of A is one fourth of the whole. Credit B only.

Solution (b) — instructor-judged. A full-credit explanation says the pieces of model A are *not all the same size*, so counting four pieces is not enough to make fourths. Model answer: "A has four pieces, but they are different sizes. Fourths have to be four equal pieces, so the shaded piece of A is not one fourth." Answers that only say "it looks wrong" or that only count the pieces do not earn full credit; the idea being graded is *equal* parts.

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6. Two ribbons of the same length are cut into 3 and 6 equal pieces. Compare one piece of each with >, <, or =.

Solution: Both ribbons started the same length. Cutting into more pieces makes every piece smaller, so a third of the ribbon is larger than a sixth of it — the picture shows the shaded third covering exactly two of the sixths. Bigger bottom number, smaller piece.

$\boxed{\frac{1}{3} > \frac{1}{6}}$

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7. Three flags of the same size are cut into 2, 4, and 8 equal pieces. Write the three shaded unit fractions in order, smallest first.

Solution: The flags are the same size, so compare the number of pieces each was cut into. Cutting into 8 gives the smallest piece, then 4, then 2: more pieces means each piece is smaller. Ordering the unit fractions smallest first therefore means ordering the bottom numbers largest first. $\boxed{\frac{1}{8}, \frac{1}{4}, \frac{1}{2}}$

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8. A garden bed is divided into 8 equal plots. (a) What unit fraction is one plot? (b) Shade three plots and name the fraction shaded.

Solution (a): The whole bed is 8 equal plots, so one plot is one eighth of the bed. $\boxed{\frac{1}{8}}$

Solution (b) — instructor-judged. The student should shade exactly three of the eight equal plots (any three) and then say the shaded amount is three copies of one eighth — three eighths. Model answer: "I shaded three plots. Each plot is $\frac{1}{8}$, so three of them is $\frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{8}$." Full credit needs both the correct shading and the counting-of-unit-fractions idea; naming three eighths with no reference to the one-eighth pieces is partial credit.